



A novel video magnification method for revealing subtle signals based on adaptive Laplacian pyramid

Yuanzhao Yang, Qi Jiang^{*}, Masood Danish

School of Control Science and Engineering, Shandong University, Jinan, 250061, China

ARTICLE INFO

Keywords:

Eulerian video magnification (EVM)
Adaptive Laplacian pyramid
Magnification factor
Subtle signals

ABSTRACT

Numerous difficult to perceive minor signals, such as minute color variations and modest movements, carry crucial information. The ability to see small signals in video has been demonstrated to be greatly aided by video magnification techniques. The Eulerian video magnification (EVM) method produces excellent results when magnifying minute colors and motions; but, it is susceptible to artifacts and outcomes that are blurry, especially when bigger magnifications are used. In this research, we propose a novel video magnification method based on adaptive Laplacian pyramid to achieve non-contact optical measurements. Experiments were conducted on real video sequences containing biological or mechanical motion signals, the results demonstrated that our approach achieves video amplification and extraction of subtle signals, supports larger amplification factors, and obtains sharper video quality with reduced artifacts and blurring.

1. Introduction

Due to physical constraints, the human eye is unable to perceive subtle signals in the world that hold crucial information [1–3]. For instance, the blood vessels during a heartbeat cause periodic changes in the color of the face [4,5] with regular beating of the pulse [6]. Buildings [7], bridges [8] and mechanical construction [9–11] generate small vibrations during operation. These tiny color changes and slight motion signals can be used for human remote monitoring of health [12,13] and diagnosis of faults in engineering [14]. However, neither are easily detectable.

In order to represent these subtle signals, methods of video magnification are investigated and can be categorized into two main types: Lagrangian perspective [15,16] and Eulerian perspective [17–21]. For the Lagrangian perspective, video magnification is achieved by tracking the trajectory of particle motion, which is computationally complex and requires detailed motion estimation and tends to introduce artifacts and noise. For the Eulerian perspective, the Eulerian magnification method is proposed to magnify tiny color changes and slight motions that are difficult to detect in real world [17]. The Eulerian video magnification (EVM) method based on principal component analysis reduces noise by decomposing the frame to find the component that best matches the small change to magnify, but the degree of motion magnification is reduced [6]. The phase-based video motion magnification method amplifies temporal multiresolution and multi-orientations local phase variations [18]. Video acceleration magnification [19] and video jerk-aware magnification methods [20] are also based on phase

magnification, which allows a better effect of slight motion magnification under large motion disturbances, but the algorithms are complex with long computation times, and require the establishment of complex multi-scale pyramids, these methods can only be applied to the magnification of motion, and are powerless for tiny color changes. In recent years, video magnification methods based on depth information fusion [22] and background segmentation have emerged to improve the quality of video magnification by pre-processing the video and separating the target region from the background motion [23,24], but they require advance delineation of the region of interest (ROI), input of more parameters or preparation of the depth camera environment.

Our work is inspired by EVM [17], a method that shows impressive results by designing spatial and temporal filters to handle tiny color or motion changes at fixed locations in the video. The algorithm of EVM is concise and efficient, and can be run in real time [25]. Unfortunately, linear EVM is sensitive to noise, especially when choosing larger magnification factors, the noise will follow the magnification.

To counter these issues, this paper proposes a multi-scale adaptive video magnification method based on adaptive Laplacian Pyramid, which optimizes the magnification factor at different spatial scales. The proposed method improves EVM [16] in two aspects: (a) The parameters such as magnification factor and spatial wavelength are obtained adaptively at different scales of the Laplacian pyramid decomposition; (b) Magnifying the motions in a larger magnification factor and reducing a significant amount of noise and blur as well as shortening the runtime.

^{*} Corresponding author.

E-mail addresses: 202020584@mail.sdu.edu.cn (Y. Yang), jiangqi@sdu.edu.cn (Q. Jiang), danish.masood.fast@gmail.com (M. Danish).

The main contributions of this paper are as follows: (a) We explore EVM theoretical methodological process to investigate the limits of magnification displacement at different scales of the Laplacian pyramid; (b) We propose a factor adaptive method based on the modified sigmoid function for each spatial frequency bands of the pyramid; (c) We use regionally averaged grey-scale extraction methods to achieve the acquisition of motion information in the ROI; (d) We demonstrate through real video sequence processing experiments that our method is faster and supports larger magnification while reducing noise and blur when compared to previous methods.

The rest of the paper is organized as follows. Section 2 describes the proposed adaptive Laplacian pyramid Euler video magnification method, Section 3 discusses the experimental results with discussion, conclusions and outlook are provided in Section 4.

2. Proposed work

In this section we introduce the theoretical foundations of EVM, and a mathematical model of EVM process based on an adaptive Laplace pyramid is developed. Finally we propose a theoretical method for adaptively optimizing the pyramid parameters.

2.1. EVM fundamentals

EVM [17] takes the first-order Taylor approximation to magnify the variation of temporal intensity at a pixel, it is able to magnify and reveal subtle signal at high spatial frequency without explicit estimation. To briefly describe the EVM, a simple case of one-dimensional translational motion signal $L(m, t)$ at position m and time t is taken as the analytical example.

A displacement function $W(t)$ is used to express the translational motion, the one-dimensional translational motion signal is described as:

$$\begin{cases} L(m, t) = f(m + W(t)) & t > 0 \\ L(m, 0) = f(m) & t = 0 \end{cases} \quad (1)$$

By using the first-order Taylor approximation, the image intensity at time t can be written as:

$$L(m, t) \approx f(m) + W(t)f'(m). \quad (2)$$

Before motion magnification, the motion at the frequency of interest is temporally filtered, and the bandpassed signal is marked by $\psi(m, t)$. If the displacement signal $W(t)$ to be amplified is in the band of the bandpass filter, then we have:

$$\psi(m, t) = W(t)f'(m). \quad (3)$$

In EVM, the bandpassed signal $\psi(m, t)$ is amplified by α . Then, this amplified signal is added to the original image intensity $L(m, t)$, as shown as:

$$\tilde{L}(m, t) = L(m, t) + \alpha\psi(m, t), \quad (4)$$

where $\tilde{L}(m, t)$ represents the output signal intensity after magnification.

Combining the above equations, it can be expressed as:

$$\tilde{L}(m, t) \approx f(m) + (1 + \alpha)W(t)f'(m). \quad (5)$$

Based on the first-order Taylor assumption, the magnification of the time-band communication signal can be correlated with the magnification results of the motion. The output signal is generated as follows:

$$\tilde{L}(m, t) \approx f(m + (1 + \alpha)W(t)). \quad (6)$$

Fig. 1 describes the principle of motion magnification using Eulerian approach for the case of a single sinusoid. In fact, the analytical process can also be applied to deal with the translational motion in two-dimensional data (i.e. video frames). In the procedures of space-time video processing, the input consecutive frames are firstly decomposed into different spatial frequency bands. Then, spatial frequency bands are temporally filtered and amplified.

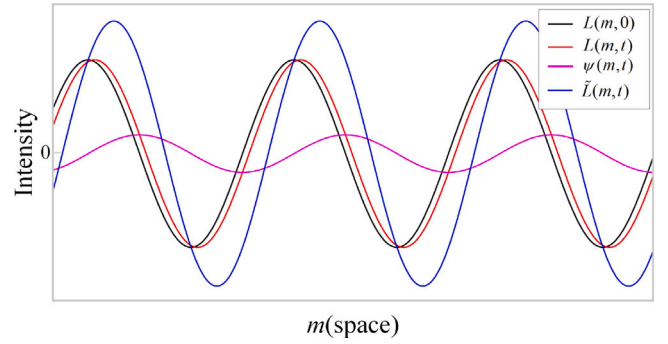


Fig. 1. Mathematical representation of EVM for a single sinusoid.

2.2. EVM based on adaptive Laplacian pyramid fundamentals

Our proposed method is shown in Fig. 2, it combines the spatial and temporal processing methods to reveal subtle signals in the video. First, each frame of the video sequence is decomposed into different spatial bands through the adaptive Laplacian pyramid. Then, we perform band-pass filtering for each spatial frequency band and select two first-order low-pass infinite impulse response (IIR) filters to construct an IIR band-pass filter to extract the frequency band of interest. After that, we multiply the extracted frequency bands of interest by magnification factors of the current level, the magnification factors for each level of the adaptive pyramid can be determined in next section. Finally, the amplified signals are added to the original spatial frequency band, and the adaptive Laplacian pyramid is reconstructed to obtain the final output video. Since our spatiotemporal filtering is performed uniformly over the pixels, the temporal and spatial consistency of the video is preserved through our video magnification method.

The magnified video is defined as $\mathbf{V} = \{I(\mathbf{x}, t) | t = 1, 2, \dots, T\}$, where $\mathbf{V}$ indicates the video, $I(\mathbf{x}, t)$ represents the frames, and $\mathbf{x} = [x, y]$ is the coordinate of the pixel. The dimensions of the frame are $M \times H$ pixels, t is the frame number, and T denotes the number of frames.

First, we build a gaussian pyramid from each frame of the video, each frame is convolved with a Gaussian convolution kernel 5×5 and down sampled to construct a Gaussian pyramid G_l from bottom to top:

$$G_l(i, j) = \sum_{m=-2}^2 \sum_{n=-2}^2 \omega(m, n) G_{l-1}(2i + m, 2j + n), \quad (7)$$

$$1 \leq l \leq N, 0 < i < C_l, 0 < j < R_l$$

where, (i, j) is coordinates of columns and rows of a point of the image at the Gaussian pyramid, N is the total number of levels of the pyramid, C_l , R_l is number of columns and rows of sub images at the l -level of the Gaussian pyramid; $\omega(m, n)$ is a 5×5 Gaussian convolution kernel, it can be given by:

$$\omega = \frac{1}{256} \begin{bmatrix} 1 & 4 & 6 & 4 & 1 \\ 4 & 16 & 24 & 16 & 4 \\ 6 & 24 & 36 & 24 & 6 \\ 4 & 16 & 24 & 16 & 4 \\ 1 & 4 & 6 & 4 & 1 \end{bmatrix}. \quad (8)$$

The Laplacian pyramid is obtained by comparing the differences between two layers of images in a Gaussian pyramid. To obtain the differences between the two layers of images, it is necessary to upsample the low-resolution image to make it the same size as the high-resolution image G_l^* . After the up sampling and Gaussian convolution of the Gaussian pyramid $l+1$ -level, it is compared with the Gaussian pyramid l -level image to obtain the l -level of the Laplacian pyramid LP_l :

$$G_l^*(i, j) = 4 \sum_{m=-2}^2 \sum_{n=-2}^2 \omega(m, n) G_{l+1}^*\left(\frac{m+i}{2}, \frac{n+j}{2}\right), \quad (9)$$

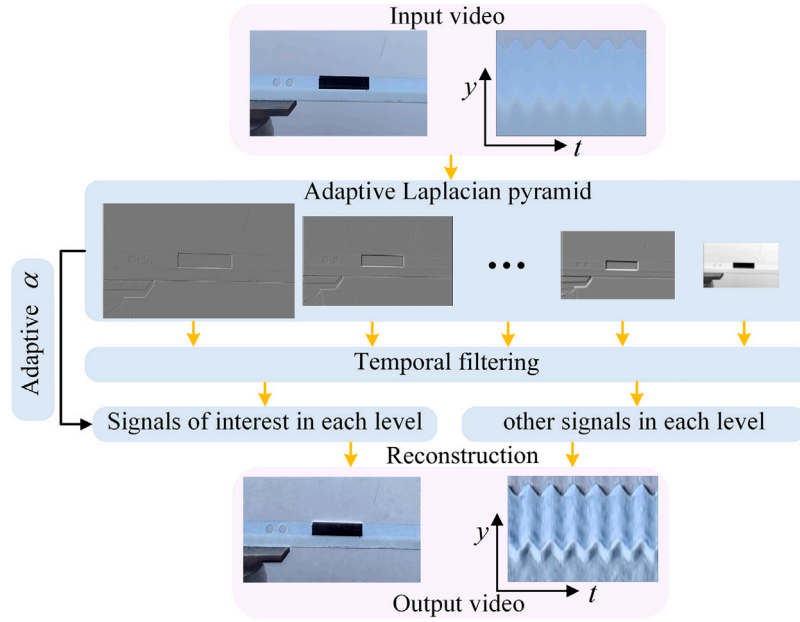


Fig. 2. Block diagram of the proposed framework.

$$\begin{cases} LP_l = G_l - G_l^*, N > l \geq 0, \\ LP_N = G_N \end{cases} \quad (10)$$

where:

$$G'_l \left(\frac{m+i}{2}, \frac{n+j}{2} \right) = \begin{cases} G_l \left(\frac{m+i}{2}, \frac{n+j}{2} \right) & \frac{m+i}{2}, \frac{n+j}{2} \in \text{int} \\ 0 & \text{other} \end{cases} \quad (11)$$

The obtained Laplacian pyramid image is filtered by time-domain band-pass filter to extract the change signal in the temporal frequency band of interest. The temporal bandpass filtering *BPF* could be implemented by the difference of two IIR low-pass filters. Next, we use an magnification factor to amplify the signal and obtain the magnified Laplacian pyramid image, it can be expressed as:

$$LP_l^{\text{mag}} = \alpha_l (BPF \otimes LP_l), \quad (12)$$

where, LP_l^{mag} and α_l are the magnified level l Laplacian pyramid image and the adaptive magnification factor, respectively. $\otimes$ is a convolution operator.

At the completion of the Laplace pyramid spatial decomposition, each level has adaptively obtained an magnification factor α_l . In the next section, we will explain the method of obtaining α_l . The reconstruction of each level Gaussian pyramid image can be expressed as:

$$G_l^{\text{mag}} = LP_l^{\text{mag}} + G_l^*, \quad (13)$$

where, G_l^{mag} is the magnified Gaussian pyramid image of level l .

Finally, we reconstruct the Gaussian pyramid and get the amplified image.

2.3. Adaptive strategy parameters

The adaptive Laplacian pyramid strategy means that the appropriate magnification factor α_l , the space wavelength λ_l , and the maximum displacement $W(t)$ are adaptively selected according to the spatial frequency of each level of the Laplacian pyramid.

This strategy has two main purposes. First is to ensure that the first-order Taylor series expansion can better approximate the subtle changes in signals, because the first-order Taylor series expansion has less error when approximating image signals with lower spatial frequencies, while for images with higher spatial frequency, the error

is high. Secondly, the signal-to-noise ratio (SNR) of images at different spatial frequencies is different. As the spatial frequency increases, the SNR will decrease.

The sigmoid function is a common function in biology, also known as the sigmoid growth curve. In this paper, we construct a modified sigmoid function to adaptively map the magnification factor to the range of 0 to α_0 . With the reduction of the spatial frequency, the magnification factor is adaptively increased.

The sigmoid function $S(x)$ can be expressed as:

$$S(x) = \frac{1}{1 + e^{-x}}, \quad (14)$$

Where e is the natural constant.

The curve transition of the sigmoid function is smooth, and the symmetry at both ends is approximately equal to the extremum. Based on this characteristic, we propose an adaptive optimization method based on The sigmoid function. The ideal magnification factor α_0 and the spatial wavelength λ_0 are first preset, and then the magnification factor of each level will be calculated as follows:

$$\alpha_l = \frac{\alpha_0}{1 + \exp\left(\frac{-2pl}{N} + p\right)}, \quad (15)$$

where, α_l is the l -level magnification factor, N is the total number of pyramid levels, l is the Laplace Pyramid level, p is a constant, this paper takes 4.

3. Experimental results and discussion

We conducted experiments using the monocular real video in the presence of biological and mechanical signals, experimental results were generated using a non-optimized MATLAB code on a desktop computer with AMD Ryzen 5 4600H CPU and 16 GB RAM. We implemented these methods in MATLAB 2020a version.

To more accurately assess how well our adaptive optimization approach using pyramidal coefficients works, we designed a comparison experiment with a linear decay amplification factor α^* and a linear stepwise decay interval of $(0.1\alpha_0, \alpha_0)$. The experimental parameters for the videos are given in Table 1. Considering that the preset amplification factor in our method is the maximum value and gradually decays, and the amplification factor in EVM is fixed at 50, we chose a larger preset amplification factor 55 to counteract the impact of this phenomenon,

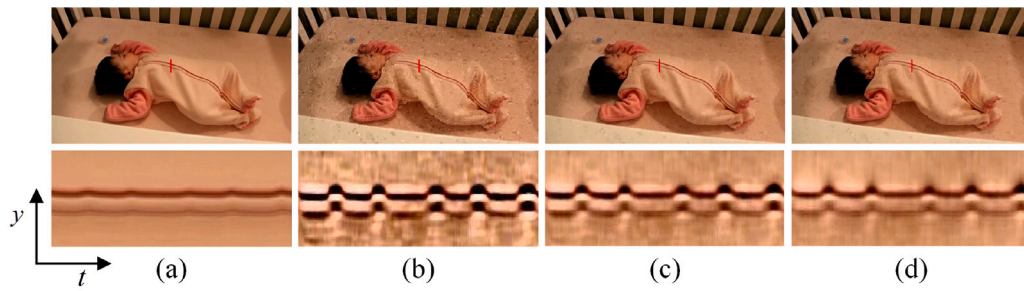


Fig. 3. Comparison of the motion magnification results on baby sequences. Top: an original frame from video. Bottom: a spatiotemporal slice along red line marked on the frame. (a) Source, (b) EVM, (c) Ours 1, (d) Ours 2.

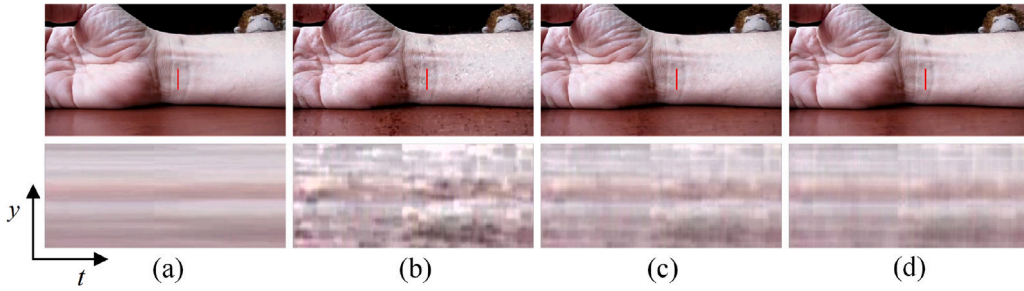


Fig. 4. Motion magnification results of slight motions of blood vessels. (a) Source, (b) EVM, (c) Ours 1, (d) Ours 2.

Table 1

Parameters for all videos.

Video	α_0 (EVM)	α_0 (Ours 1)	α_0 (Ours 2)	λ	FPS
Video1	50	55	55	4	30
Video2	50	55	55	16	30
Video3	50	55	55	4	30

Table 2

Temporal frequency.

Pyramid level	EVM	Ours 1	Ours 2
8	50	55	54.01
7	50	48.81	52.39
6	50	42.62	48.44
5	50	36.44	40.21
4	50	30.25	27.5
3	50	24.06	14.8
2	50	17.88	6.56
1	50	11.69	2.61

in ours2, each layer amplification factor is obtained by inputting a predetermined 55 to the adaptive amplification factor equation. We provide representative videos and results to describe our amplification method, where the video frames are processed in Luminance, In-phase, Quadrature (YIQ) color space and on each channel independently. For simplicity of reading, the linear attenuation factors method for the control is denoted by ours 1 and the modified sigmoid function factor adaptive method by ours 2.

The preset amplification factor for each video is 55, the differences in video size $M \times H$ and λ result in different pyramid layers after video frame decomposition, and the value and number of adaptive amplification factors are also different. Taking video 1 as an example, the number of decomposed pyramid layers is 8. Table 2 shows the comparison of the magnification factors in the pyramid between [17] and this scheme.

3.1. Video experiment

We chose two video sequences 1, 2 in [17] with biological signals as input and the magnification factor was set to 50 for the comparison

of video magnification processing effects, the source videos 1 and 2 were recorded at a frame rate of 30 frames per second using a standard point-and-shoot camera. The first video shows the undulation of the chest during the breath of a newborn, as shown in Fig. 3(a). Its motion is amplified by the original EVM, factor adaptive through linear attenuation or modified sigmoid function as shown in (b) and (c), (d), respectively. Considering that every act of breathing consists of two main movements: inhalation and exhalation, it is through the movement of the thorax that we can obtain the respiratory signals. By comparing slices in the y -direction of the thoracic heaving area with time t , the Respiratory signal implied by baby's chest was successfully extracted. Referring to Figs. 3 and 14, the noise in EVM method video is the most pronounced, and the lowest noise level was observed in ours 2. After the newborn's chest position is enlarged by EVM, most of the pixel intensities are close to the extreme values of 0 and 255, losing much edge and color information, but the details of ours 1 and ours 2 remain better expressed.

Fig. 4(a) shows frames from video 2 with the slight motion of the adult right wrist vessels, and the magnified pulsatile arteries become clearer when observing the spatiotemporal slices from inputs (b), (c) and (d). It can be observed that our method supports larger magnification and significantly reduces artifacts and noise.

A bench-scale model of an aluminum beam structure was used to validate the proposed method in video 3. Fig. 5 shows the overall experimental setup including a signal generator (DG1022U) and a power amplifier (DTCLA-200), the aluminum beam structure was fixed on a shaker table. A sinusoidal signal with a frequency of 1 Hz was generated to vibrate the aluminum beam. The structure was recorded at a frame rate of 30 frames per second using an iPhone 11's camera, the straight-line distance between the vibrator and the camera is about 350 mm, and the lighting environment was normal indoor lighting conditions without any external lighting enhancement. We down-sample the pixels to a resolution of 460×760 . As shown in Fig. 6, the vibration signals of the aluminum beam structure are effectively amplified at a large magnification, our method reduces artifacts and noise generation.

3.2. Extraction of subtle signals

The Eulerian view of the video magnification process enhances the range and intensity of pixel grey scale and color changes in motion

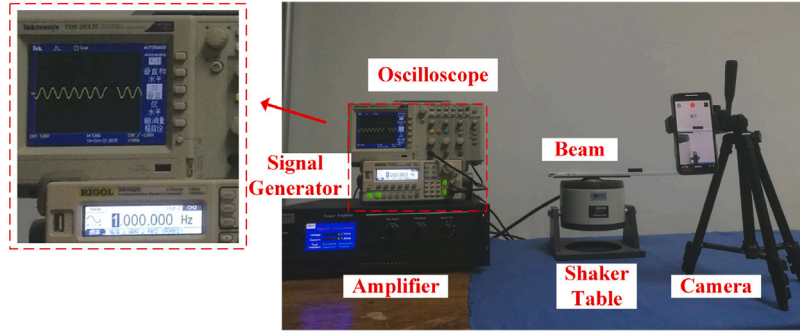


Fig. 5. The experimental setup.

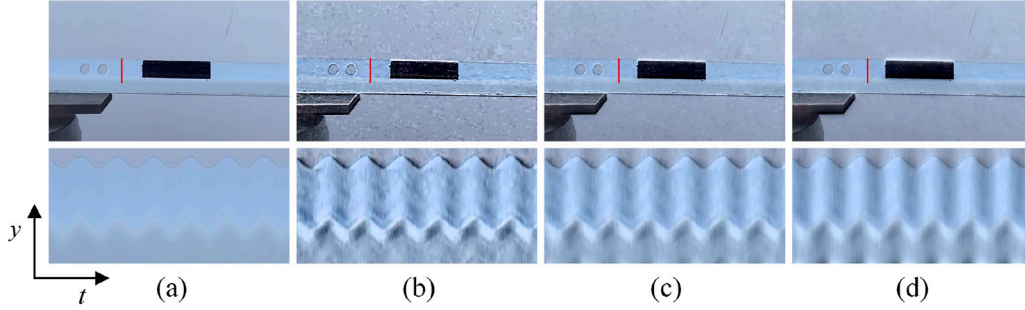


Fig. 6. Results of magnifying the vibration signals video of an aluminum beam structure. (a) Source, (b) EVM, (c) Ours 1, (d) Ours 2.

regions, and motion changes are more closely related to the intensity of grey scale between video frames, so we used motion region grey scale correlation to measure small movements in the video. In order to improve the accuracy of motion signals extraction, we use the method of extracting the average value of the gray-scale region $\bar{I}(x, y)$. For a given coordinate point (x, y) , region size $M \times N$, it can be obtained by the following equation.

$$\bar{I}(x, y) = \frac{1}{M \times N} \sum_{I=1}^{M-1} \sum_{J=1}^{N-1} I(x + I, y + J), \quad (16)$$

where I and J are the horizontal and vertical distances of points in the region of interest from the given coordinate point, respectively.

the ROI of size 50×50 in video 1, 2, 3 are selected at a suitable location, and the average value of the gray-scale region is extracted as the signal of interest. As shown in Figs. 7 to 12, the biological signals and mechanical vibration signals in the original video are relatively subtle and difficult to find. After motion magnification process, the signals can be seen as obvious periodic peaks, and are consistent with the real respiratory rate, pulse signal frequency, and beam vibration signals. Ours 2 retains more details than ours 1, as shown in Fig. 12, with smoother peaks. In Fig. 12, the range of intensity of the original motion signal is ± 0.014 . The range of intensity of the amplified motion is ± 0.255 , with an increased range of intensity variation. The average intensity of the original motion curve is 0.6911, the average intensity value for each vertex of the EVM method is 0.6863, and the average intensity value for each vertex of the ours 1 is 0.6864, and this of the ours 2 is 0.6872, indicating to some extent that our method is closer to the original motion signal. The results demonstrate that our research is effective for amplifying subtle signals.

After obtaining the motion signals of the motion regions, we performed experiments to measure the motion frequencies. Fig. 13 shows the results of the Fast Fourier Transform (FFT) of the extracted motion signals. Baby respiratory frequencies were shown from 0.3 to 1 Hz, corresponding to 18 to 60 respirations per minute. The adult pulse experiment used a frequency of 0.7 to 1.5 Hz, corresponding to a pulse rate of 42 to 90 beats per minute in the relaxed state. Beam vibration

was set to 0.5 to 1.5 Hz due to the standard 1 Hz vibration chosen for the experiment interval. We obtained the maximum amplitude of the baby's breath at 0.5357 Hz, pulse at 0.9268 Hz, beam at 1 Hz, respectively.

The respiratory rate is obtained by observing the cycle of the rise and fall of the baby's chest when breathing 4.25 chest rises and falls every eight seconds in video 1 (there is a rise of the chest to the highest point at the end of the video, which we count as 0.25 chest rises and falls). This can be corresponded to by the peak in the movement curve Fig. 8. The respiratory rate is 0.5312 Hz. Our measured peak frequency is 99.15% accurate to the actual respiratory rate.

The frequency of the pulse in the original video is obtained by the undulation of the radial artery in the video, which is 0.931 Hz, and the peak frequency we measured is 0.9268 Hz, which is 99.54% accurate.

The frequency of the beam vibration video can be obtained from the information displayed on the signal generator and oscilloscope screen as shown in Fig. 5 as 1 Hz and we measured a peak frequency of 1 Hz with 100% accuracy.

3.3. Video quality evaluation

Improvements in output image quality will enable more precise measurements for any application reliant on magnified video sequences, and may enable further development of video motion magnification techniques in new technical fields.

We have chosen video quality metrics structural similarity (SSIM), peak signal to noise ration (PSNR), root mean square error (RMSE) to quantitatively compare the quality of motion amplified videos [26,27]. SSIM is obtained by calculating the similarity to the original image in the three modules luminance, contrast and structure, PSNR is obtained by calculating, with ten as the base image YIQ space, the square of the intensity maximum (255) of the pixels in the Y channel divided by the logarithm of the value of the mean square error (MSE) and multiplied by ten, and RMSE is the square root of MSE. The SSIM, PSNR, and RMSE values of each image frame of the video are calculated and then averaged to obtain the objective evaluation index parameter values of the video.

Table 3

The quality assessment results of video sequences.

Mi(SSIM, PSNR, RMSE)	EVM	Ours 1	Ours 2
video 1	M1(0.94,24.84,7.83)	M2(0.98,29.63,5.93)	M3(0.99,30.95,5.55)
video 2	M4(0.78,23.21,7.61)	M5(0.92,28.29,5.61)	M6(0.95,30.01,4.99)
video 3	M7(0.71,19.51,7.16)	M8(0.91,20.54,6.31)	M9(0.91,20.67,6.22)

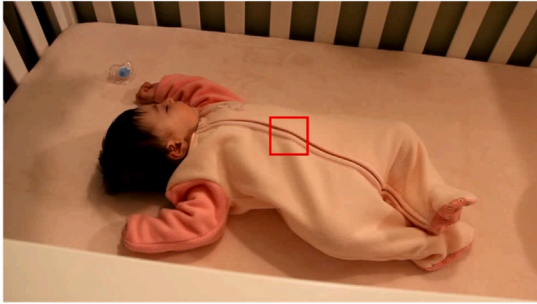
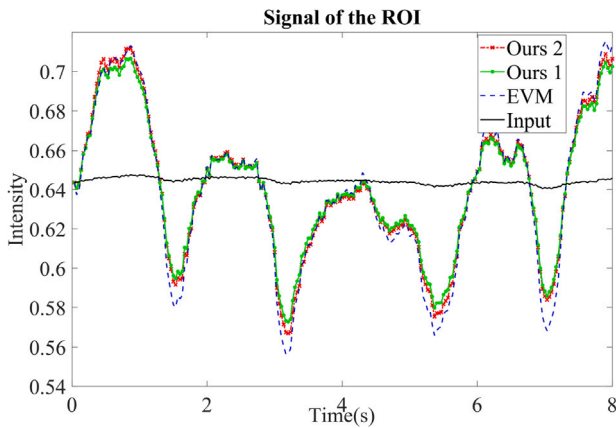
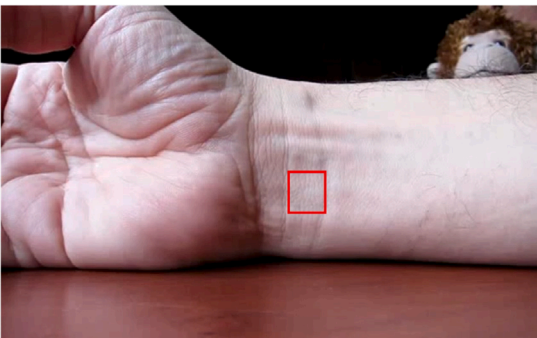
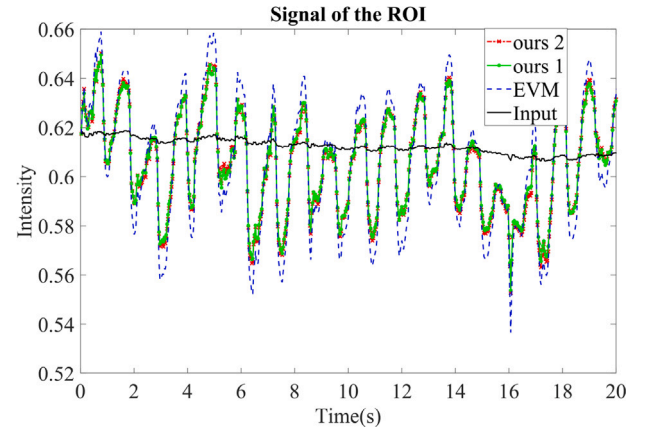
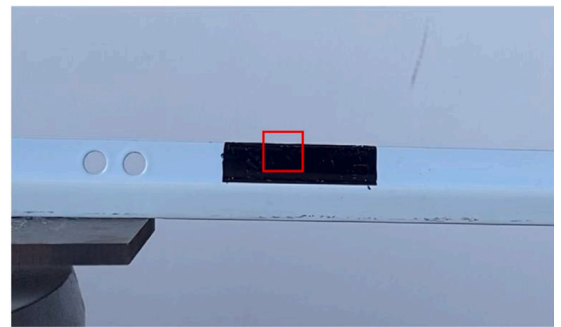
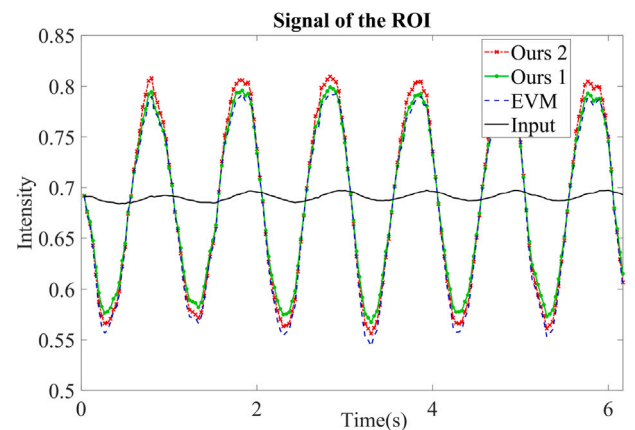
**Fig. 7.** A frame from video 1, the ROI is indicated by the red box.**Fig. 8.** Signal of the ROI in video 1.**Fig. 9.** A frame from video 2, the ROI is indicated by the red box.**Fig. 10.** Signal of the ROI in video 2.**Fig. 11.** A frame from video 3, the ROI is indicated by the red box.**Fig. 12.** Signal of the ROI in video 3.

Table 3 shows the SSIM, PSNR, RMSE average values of all frames of video amplified videos generated with our method and [17]. The quality of the images increases with SSIM. As shown in Figs. 14 to 16, compared to EVM, our method improves the value of SSIM for each frame, indicating a higher overall structural similarity to the original video, reducing the generation of noise and significantly improving the quality of the video.

The video motion magnification process introduces a different quantity of noise on each frame, which generates fluctuations in the SSIM.

The fluctuations are more pronounced in Fig. 15 because video2 is 20 s long and contains more information about the frames. The fluctuation interval of the video frame SSIM values obtained by our method is smaller, which indicates a more stable quality of the video frames.

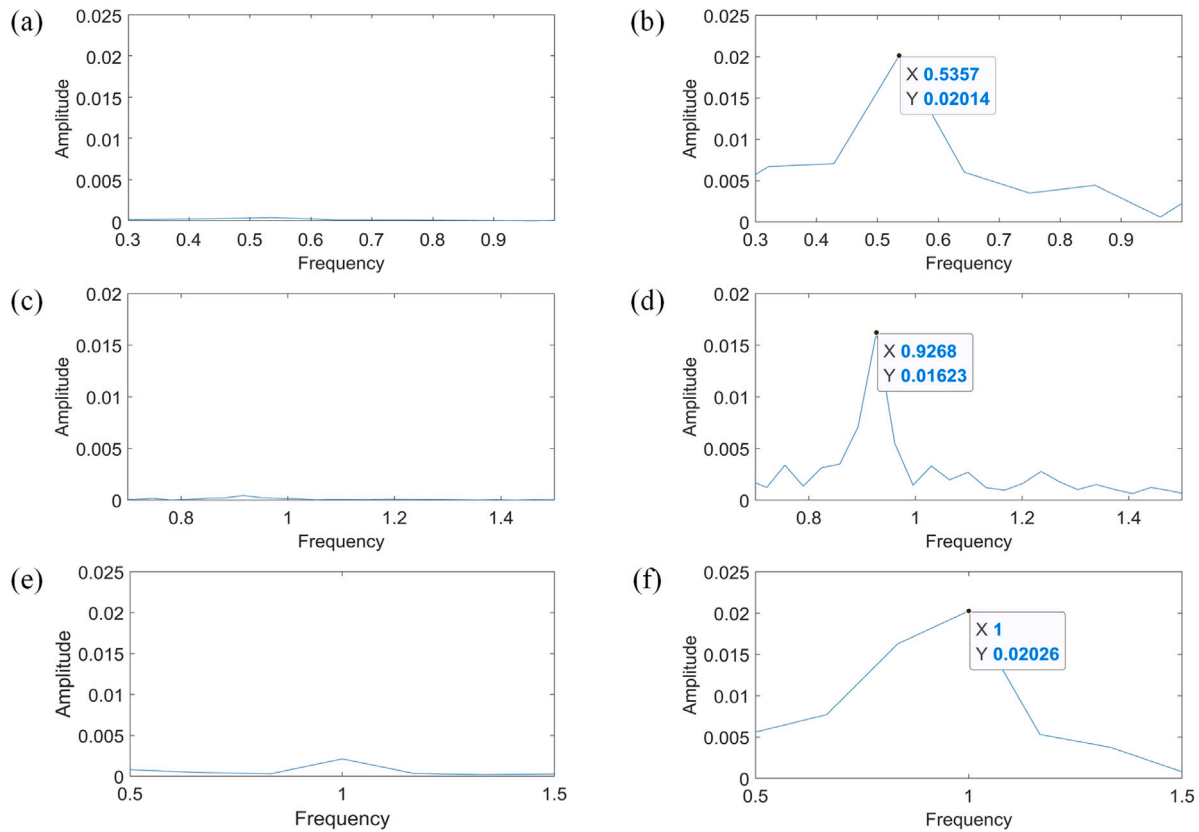


Fig. 13. FFT of motion signals. (a) Video 1, (b) Video 1 processed by ours 2, (c) Video 2, (d) Video 2 processed by ours 2, (e) Video 3, (f) Video 3 processed by ours 2.

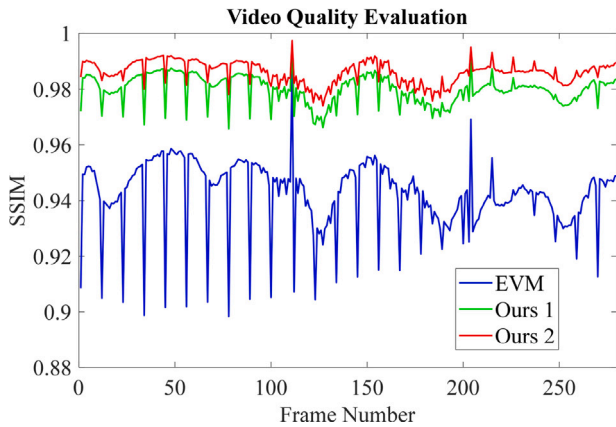


Fig. 14. SSIM results of video 1.

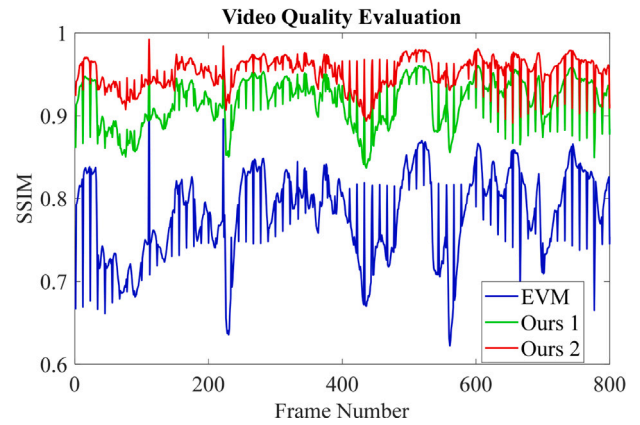


Fig. 15. SSIM results of video 2.

Both PSNR and RMSE measure image quality by calculating the global magnitude of the pixel error between the image to be evaluated and the reference image; A larger PSNR value indicates less distortion between the image to be evaluated and the reference image, and better image quality. Our method obtains better PSNR and RMSE metrics, indicating a reduction in image distortion and an improvement in image clarity. Table 3 shows that the proposed method has better video quality compared to EVM. Ours 2 provides better video quality evaluation metrics than ours 1 and EVM, proving that the proposed method of adaptive selection of amplification factors based on sigmoid function is better than the linear attenuation amplification factor selection method. This is attributed to the smoothness of the improved sigmoid function, as it tends gently to the extremes at both ends.

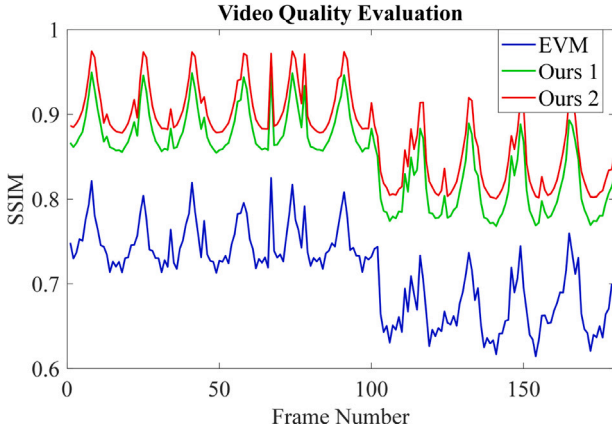
We conducted experiments and obtained the video quality evaluation indicators of the ours 2 method with a preset amplification factor of 50; abbreviations: ours 2 (50), the SSIM, PSNR and RMSE indices of ours 2 (50) with values of (0.99, 31.74, 5.15), which are better than EVM (50) and better than ours 2 (55). By comparing ours 2 (50) and ours 2 (55), we found that when choosing a larger amplification factor, the noise of the video frame increases and the overall quality of the video decreases. By comparing EVM (50) with ours 2 (55), it is demonstrated that our method actually improves video quality when selecting a larger amplification factor, rather than reducing it.

Table 4 shows the motion magnification execution time. Our method reduces the time required for the video magnification process, which improves the efficiency of the calculation.

Table 4

Running times of different methods (in seconds).

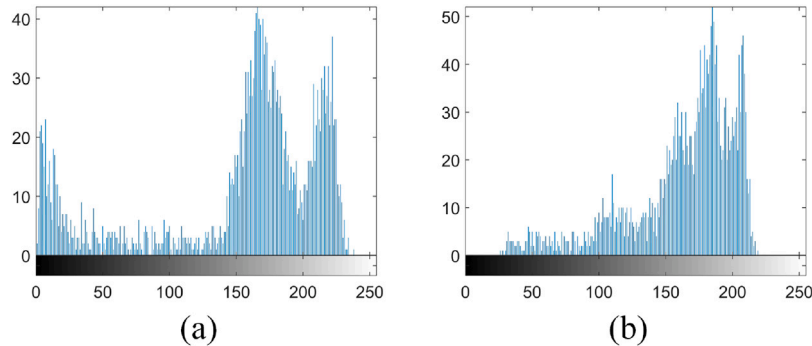
Video	Resolution ($H * M * t$)	EVM	Ours 1	Ours 2
Video1	544 * 960 * 281	51.73	51.55	51.25
Video2	352 * 640 * 874	77.22	65.46	64.6
Video3	460 * 760 * 190	27.63	23.74	23.63

**Fig. 16.** SSIM results of video 3.

We performed a comparison experiment of the greyscale intensity distribution of all pixels in the motion magnification region of interest, as shown in Fig. 17. EVM produces pixel points near the extremes 0 and 255 in the motion region, which causes distortion in the motion region, losing some detailed information about the motion, and the noise grows linearly as the magnification factor increases, causing relatively larger distortion. However, our method produces almost no pixel points near the extremes, with pixel intensities ranging between 25 and 225, meaning that our method can amplify tiny motions in the video while retaining more detailed information at large amplification factors over 50, making it more suitable for motion analysis work.

4. Conclusions and future work

To achieve non-contact optical measurements, we propose a multi-level optimized factor video magnification method based on adaptive Laplacian pyramid. In this method, the magnification factor is adaptively selected according to the spatial frequency of each pyramid levels. The lower frequency bands of the adaptive Laplacian pyramid support larger magnification factors.

**Fig. 17.** Comparison of histograms of pixel intensity distribution for motion areas in video1. (a) EVM, (b) Ours 2.

By applying several video quality evaluation metrics, it is demonstrated that the video processed by our method has better quality and improves efficiency of algorithm operation. The strategy solves the problem of generating more artifacts and noise at large magnification factor in [17]. Therefore, the method is available for subtler signal magnification. Other suitable objective metrics can be investigated in the future, and these can be used to provide additional perspectives to verify the effectiveness of video amplification methods.

Our method effectively extracted the amplified signals of infant respiration, pulse and beam vibration by extracting the average value of the gray-scale region, which can be well observed in the processed video by human eyes. The results demonstrate that our proposed method has a wide range of application prospects.

This method inherits the advantages of EVM, such as removing the need to manually annotate regions of interest, and no tracking or region alignment. However, it also has some limitations, one is the assumption of linear motion, which is not effective for nonlinear small changes and still has room for improvement, and the other is the need for a stable video shooting environment, where camera shake and large motions in the background can adversely affect the magnification of subtle signals. Future strategy involves devising more stable methods to magnify the subtle signals.

CRediT authorship contribution statement

Yuanzhao Yang: Conceptualization, Methodology, Validation, Writing – original draft, Review & editing. **Qi Jiang:** Funding acquisition, Project administration. **Masood Danish:** Review, Proofreading & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This work was supported in part by the National Natural Science Foundation of China under Grant 61973194, in part by the Major Scientific and Technological Innovation Project of Shandong Province, China under Grant 2019JZZY010427 and in part by the Shenzhen Fundamental Research and Discipline Layout Project, China under Grant JCYJ20190806155616366.

References

- [1] D. Alinovi, L. Cattani, G. Ferrari, et al., Spatio-temporal video processing for respiratory rate estimation, in: IEEE International Symposium on Medical Measurements and Applications (MeMeA) Proceedings, 2015, pp. 12–17, <http://dx.doi.org/10.1109/MeMeA.2015.7145164>.
- [2] A. Alzahrani, J. Hosseinkhani, S. Rajan, et al., Reducing motion impact on video magnification using wavelet transform and principal component analysis for heart rate estimation, in: IEEE International Instrumentation and Measurement Technology Conference, I2MTC, 2021, pp. 1–6, <http://dx.doi.org/10.1109/I2MTC50364.2021.9460026>.
- [3] E. Perez, E. Zappa, Video motion magnification to improve the accuracy of vision-based vibration measurements, IEEE Trans. Instrum. Meas. 71 (2022) 1–12, <http://dx.doi.org/10.1109/tim.2022.3175977>.
- [4] M.Z. Poh, D.J. McDuff, R. Picard, Non-contact, automated cardiac pulse measurements using video imaging and blind source separation, Opt. Express 18 (10) (2010) 10762–10774, <http://dx.doi.org/10.1364/OE.18.010762>.
- [5] Z. Yang, X.Z. Yang, J. Jin, et al., Motion-resistant heart rate measurement from face videos using patch-based fusion, Signal Image Video Process. 13 (3) (2019) 423–430, <http://dx.doi.org/10.1007/s11760-018-01409-w>.
- [6] X. Wu, X.Z. Yang, J. Jin, et al., PCA-based magnification method for revealing small signals in video, Signal Image Video Process. 12 (7) (2018) 1293–1299, <http://dx.doi.org/10.1007/s11760-018-1282-0>.
- [7] I. Roselli, M. Malena, M. Mongelli, et al., Health assessment and ambient vibration testing of the “ponte delle torri” of spoleto during the 2016–2017 central Italy seismic sequence, J. Civil Struct. Health Monit. 8 (2) (2018) 199–216, <http://dx.doi.org/10.1007/s13349-018-0268-5>.
- [8] Z. Liu, Q. He, S. Chen, et al., Time-varying motion filtering for vision-based nonstationary vibration measurement, IEEE Trans. Instrum. Meas. 69 (6) (2020) 3907–3916, <http://dx.doi.org/10.1109/tim.2019.2937531>.
- [9] D. Siringoringo, S. Wangchuk, Y. Fujino, Noncontact operational modal analysis of light poles by vision-based motion-magnification method, Eng. Struct. 244 (2021) 112728, <http://dx.doi.org/10.1016/j.engstruct.2021.112728>.
- [10] L. Felipe-Sesé, A.J. Molina-Viedma, M. Pastor-Cintas, et al., Exploiting phase-based motion magnification for the measurement of subtle 3D deformation maps with fp+2d-DIC, Measurement 195 (2022) 111122, <http://dx.doi.org/10.1016/j.measurement.2022.111122>.
- [11] N.A. Valente, C.T. do Cabo, Z. Mao, et al., Quantification of phase-based magnified motion using image enhancement and optical flow techniques, Measurement 189 (2022) 110508, <http://dx.doi.org/10.1016/j.measurement.2021.110508>.
- [12] G.O. Ganfure, Using video stream for continuous monitoring of breathing rate for general setting, Signal Image Video Process. 13 (7) (2019) 1395–1403, <http://dx.doi.org/10.1007/s11760-019-01486-5>.
- [13] A. Alarifi, A. Tolba, A.S. Hassanein, Visualization process assisted by the Eulerian video magnification algorithm for a heart rate monitoring system: mobile applications, Multimedia Tools Appl. 79 (7–8) (2018) 5149–5160, <http://dx.doi.org/10.1007/s11042-018-6313-x>.
- [14] P. Poozesh, A. Sarrafi, Z. Mao, et al., Feasibility of extracting operating shapes using phase-based motion magnification technique and stereo-photogrammetry, J. Sound Vib. 407 (2017) 350–366, <http://dx.doi.org/10.1016/j.jsv.2017.06.003>.
- [15] G. Balakrishnan, F. Durand, J. Guttag, Detecting pulse from head motions in video, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2013, pp. 3430–3437, <http://dx.doi.org/10.1109/cvpr.2013.440>.
- [16] C. Liu, A. Torralba, W.T. Freeman, et al., Motion magnification, ACM Trans. Graph. 24 (3) (2005) 519–526, <http://dx.doi.org/10.1145/1073204.1073223>.
- [17] H. Wu, M. Rubinstein, E. Shih, et al., Eulerian video magnification for revealing subtle changes in the world, ACM Trans. Graph. 31 (4) (2012) 1–8, <http://dx.doi.org/10.1145/2185520.2185561>.
- [18] N. Wadhwa, M. Rubinstein, F. Durand, et al., Phase-based video motion processing, ACM Trans. Graph. 32 (4) (2013) 1–10, <http://dx.doi.org/10.1145/2461912.2461966>.
- [19] Y. Zhang, S.L. Pintea, J.C.V. Gemert, Video acceleration magnification, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2017, pp. 529–537, <http://dx.doi.org/10.48550/arXiv.1704.04186>.
- [20] S. Takeda, O. K., D. Mikami, et al., Jerk-aware video acceleration magnification, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2018, pp. 1769–1777, <http://dx.doi.org/10.1109/cvpr.2018.00190>.
- [21] S. Takeda, M. Isogai, S. Shimizu, et al., Local Riesz pyramid for faster phase-based video magnification, IEICE Trans. Inf. Syst. 103 (10) (2020) 2036–2046, <http://dx.doi.org/10.1587/transinf.2020PCP0006>.
- [22] J.F.P. Kooij, J.C. Gemert, Depth-aware motion magnification, in: European Conference on Computer Vision, Vol. 9912, 2016, pp. 467–482, http://dx.doi.org/10.1007/978-3-319-46484-8_28.
- [23] S. Yadav, P. Bhalkare, U. Verma, Region-based stabilized video magnification approach, in: Sustainable Advanced Computing, Springer, 2022, pp. 381–394, http://dx.doi.org/10.1007/978-981-16-9012-9_31.
- [24] M. Verma, S. Raman, Interest region based motion magnification, in: International Conference on Image Analysis and Processing, in: Lecture Notes in Computer Science, 2017, pp. 27–39, http://dx.doi.org/10.1007/978-3-319-68560-1_3.
- [25] N. Wadhwa, H.Y. Wu, A. Davis, et al., Eulerian video magnification and analysis, Commun. ACM 60 (1) (2016) 87–95, <http://dx.doi.org/10.1145/3015573>.
- [26] Z. Wang, A.C. Bovik, Image and multidimensional signal processing-a universal image quality index, IEEE Signal Process. Lett. 9 (3) (2002) 81–84, <http://dx.doi.org/10.1145/3015573>.
- [27] Sheikh, R. Hamid, F. Muhammad, et al., A statistical evaluation of recent full reference image quality assessment algorithms, IEEE Trans. Image Process. 15 (11) (2006) 3440–3451, <http://dx.doi.org/10.1109/tip.2006.881959>.